

A. Objectives

1. To prepare a nutrient-dense multi-grain flour using ragi, bajra, and maize.
2. To analyze the nutritional benefits of the composition of mixed flour
3. To determine the best composite flour ratio, grinding method, and storage technique for maximum nutrition and shelf life.
4. To develop the product based on the mixed flour
5. To create awareness among the people/ SHG to enhance their economic levels

A.1 Manpower

1. Technical person – 1
2. Non-technical person (labour)-1

A.2 Equipment's

S.No.	Equipment	quantity	Cost
1	Sieve / Cleaning Tray	5 – 6 (as per requirement)	1000
2	Sun Drying	-	
3	Stone Grinding Mill (Chakki)	1	25000
4	Weighing machine	2	10000
5	Mixing Bowl (Large)	10	4000
6	Airtight Container	12	5000
7	Refrigerator	1	30000

A.3 Raw Material/Consumables

S.No.	Grain	Common Name	Quantity Required
1	Ragi	Finger Millet	500 grams
2	Bajra	Pearl Millet	300 grams
3	Maize	Corn (Dry)	200 grams
Total			1000 grams (1 kg)

B. Introduction

In recent years, there has been a growing demand for healthy and nutrient-rich food products due to increasing lifestyle diseases and health awareness. Traditional grains like **Ragi (Finger Millet)**, **Bajra (Pearl Millet)**, and **Maize (Corn)** are highly nutritious, easily available, and offer several health benefits.

- ✓ **Ragi** is exceptionally rich in Calcium (344 mg/100g), making it the best grain for bone health.
- ✓ **Bajra** is loaded with Protein (11.6 g/100g) and Iron (8.0 mg/100g), which helps in muscle building and anemia prevention.
- ✓ **Maize** provides good Energy (342 kcal/100g), Fiber (2.7 g/100g), and natural sweetness which improves taste.

This project focuses on developing a multi-grain flour blend using these three grains in an optimized ratio. The study uses **Stone Milling** as the preferred grinding technique because it retains maximum nutrients, natural oils, and dietary fiber compared to industrial ball milling.

The goal is to prepare a flour that supports:

- Bone strength
- Muscle growth
- Digestion
- Daily energy needs

The following table shows the nutritional value of each grain per 100 grams. This data forms the scientific basis of the blend.

Nutrient (per 100g)	Ragi (Finger Millet)	Bajra (Pearl Millet)	Maize (Dry Corn)
Energy	328 kcal	361 kcal	342 kcal
Protein	7.3 g	11.6 g	11.1 g
Carbohydrates	72 g	67 g	66 g
Dietary Fiber	3.6 g	1.2 g	2.7 g
Total Fat	1.3 g	5.0 g	3.6 g
Calcium	344 mg	42 mg	10 mg
Iron	3.9 mg	8.0 mg	2.0 mg
Phosphorus	283 mg	296 mg	210 mg

Source: Nutritive Value of Indian Foods, NIN Hyderabad

After studying the nutritional profiles, the following **50:30:20 [R:B:M] ratio** is proposed as the most balanced and health-effective blend.

Grain	Percentage	Quantity (for 1 kg flour)	Primary Nutritional Role
Ragi	50%	500 grams	Calcium for Bone Health + Soluble Fiber
Bajra	30%	300 grams	Protein for Muscle + Iron for Blood
Maize	20%	200 grams	Energy + Insoluble Fiber + Natural Sweetness
Total	100%	1000 grams	

Calculated nutritional value of the composition 50:30:20 blend (per 100g)

Nutrient	Calculation	Approx. Final Value
Energy	$(328 \times 0.5) + (361 \times 0.3) + (342 \times 0.2)$	341 kcal
Protein	$(7.3 \times 0.5) + (11.6 \times 0.3) + (11.1 \times 0.2)$	9.4 g
Carbohydrates	$(72 \times 0.5) + (67 \times 0.3) + (66 \times 0.2)$	69.3 g
Dietary Fiber	$(3.6 \times 0.5) + (1.2 \times 0.3) + (2.7 \times 0.2)$	2.7 g
Total Fat	$(1.3 \times 0.5) + (5.0 \times 0.3) + (3.6 \times 0.2)$	2.9 g
Calcium	$(344 \times 0.5) + (42 \times 0.3) + (10 \times 0.2)$	186.6 mg
Iron	$(3.9 \times 0.5) + (8.0 \times 0.3) + (2.0 \times 0.2)$	4.8 mg

C. Methodology

1. Selection of Grains

- Select good quality, clean, and pest-free grains of Ragi, Bajra, and Maize.
- Ensure grains are from a fresh harvest for best nutritive value.
- Discard any discolored or damaged grains.

2. Cleaning

- Manually remove stones, dust, broken grains, and any foreign particles.
- Sieve the grains thoroughly so that no impurities remain.

3. Sun Drying (Crucial Step)

- Spread the cleaned grains on a clean, dry cotton cloth under direct sunlight.
- Dry for 4-6 hours (or until completely moisture-free).
- Moisture causes fungal growth and reduces shelf life.

4. Weighing and Measuring

Through using of digital weighing machine, measure the grains exactly as per the ratio. Ragi [500 grams], Bajra [300 grams], and Maize [200 grams].

5. Mixing the Whole Grains

- In a large container, mix all three grains together thoroughly.
- This ensures that when fed into the stone mill, the grinding happens simultaneously.
- Simultaneous grinding helps in uniform particle size and flavor blending.

I. Process:

- Feed the mixed grains slowly into the stone mill (Chakki).
- Maintain RPM (rotation per minute) to prevent heat generation.
- Collect the flour in a clean container from the outlet.

II. **Benefits of Stone Milling:**

- a. Low Heat Generation: Protects heat-sensitive nutrients like calcium and B-vitamins.
- b. Whole Grain Retention: Stone crushes the entire grain including bran (fiber) and germ (healthy oils).
- c. Natural Taste: Produces a characteristic nutty, wholesome flavor.
- d. No Chemical Additives: 100% natural process.

6. **Storage**

- a. Transfer the finished flour immediately into a clean, dry, airtight glass or BPA-free plastic container.
- b. Seal the lid tightly.
- c. Label the container with:
 - d. Product Name: Ragi-Bajra-Maize Multi-Grain Flour
 - e. Date of Preparation: [Current Date]
 - f. Ratio: 50:30:20 [R:B:M]
- g. Subsequent studies will evaluate the effect of extended storage on product integrity.

C.2 WORKING PRINCIPLE

This project is scientifically based on the principle of Nutritional Synergy.

1. Complete Protein Formation:

Ragi is deficient in the amino acid Lysine but rich in Methionine. Bajra and Maize are deficient in Methionine but contain Lysine. When combined in the 50:30:20 ratio, they complement each other to form a Complete Protein Profile similar to animal sources, which is essential for muscle building.

2. Calcium-Phosphorus Balance:

Ragi provides high Calcium (344 mg). Bajra provides high Phosphorus (296 mg). Calcium and Phosphorus work together for proper bone mineralization. One without the other is ineffective.

3. Fiber Synergy for Digestion:

Ragi contains Soluble Fiber which absorbs water and forms a gel, softening stool. Maize and Bajra contain Insoluble Fiber which adds bulk to stool and promotes bowel movement. Together, they prevent constipation effectively without causing gas or bloating when water intake is adequate.

C3. Possible value addition

The prepared multi-grain flour can be used in the following ways:

Application	Preparation Method
Multi-Grain Roti/Chapati	Mix with warm water, knead into dough, roll and cook on tawa
Health Porridge	Cook 2 tbsp flour with water/milk until thick; add jaggery or salt
Infant Food (8+ months)	Roast flour lightly, cook in water until smooth paste
Nutritional Drink (Kanji)	Mix with buttermilk, salt, and cumin; ferment overnight
Fitness Dosa	Make batter with curd, ferment 4 hours, spread on hot tawa
Ladoo/Snacks	Dry roast flour in ghee, mix with jaggery and nuts, shape into balls

DETAILED COST ANALYSIS (Approximate)

Item	Quantity	Approx. Cost (INR)
Ragi (Finger Millet)	500g	₹30 – 40
Bajra (Pearl Millet)	300g	₹20 – 25
Maize (Corn)	200g	₹15 – 20
Grinding Charges (Stone Mill)	1 kg	₹20 – 30
Airtight Container	1 piece	₹50 – 80
Total Project Cost		₹135 - ₹195

This makes it an extremely economical and affordable health food option for all families.

REFERENCES

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